

Considering minimal requirements, CloudStack deployment consists of two machines – one running the CloudStack management server and the other acting as the cloud infrastructure. Even a single machine can perform the tasks of both the management server and the hypervisor host.

Figure 1 highlights a simple installation and deployment of CloudStack. This deployment is extensible to multi-node server installations where hundreds and thousands of client nodes can be joined in dynamic fashion.

Management server: The management server is defined as a dedicated server in the network, which controls the allocation of virtual machines to hosts, and performs tasks like storage allocation and IP allocations to instances. It runs in Apache Tomcat Container and uses the MySQL database server for storage operations.

It performs the following tasks:

- Manages all virtual machine storages, backups and instances
- Is a Web interface to clients and administrators
- Monitors the resources, disk images and ISO images
- Supports single point configuration for cloud
- Offers API interface support for CloudStack API and EC2 interface

Cloud infrastructure: All the resources in the cloud are suitably

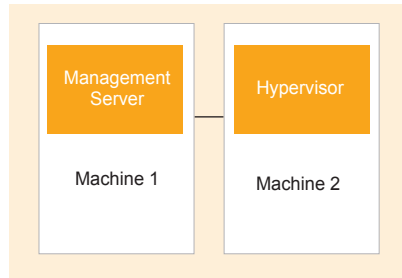


Figure 1: Simple CloudStack deployment

managed and monitored as follows.

- **Regions:** Geographically proximity zones managed by management servers
- **Zones:** Single data centre and consists of secondary storage
- **Pods:** A rack or row of racks that includes a 2-layer switch and one or more clusters
- **Host:** Client node
- **Primary storage:** Attached to single cluster running disk images
- **Secondary storage:** Stores templates, ISO images and snapshots

CloudStack: Implementation guidelines

CloudStack is an open source engineering framework for hybrid cloud development on an IaaS platform service. It can be used on public cloud like AWS, as it supports native EC2 and S3 compatible API to deploy cloud services. It can also be used with private cloud platforms or hybrid cloud solutions.

This framework is a complete package of cloud infrastructure services including containerisation (container-as-a-service), compute orchestration, network-as-a-service, account management, user management, open native API services and command line interface (CLI).

CloudStack can be used with CLI or Web application UI or dashboard services and RESTful API services. It supports all the popular hypervisors like VMware, Citrix XenServer, KVM, Xen, Cloud Platform (XCP), Microsoft Hyper-V and Oracle VM server. It manages and orchestrates pools of storage, network and computer resources to build a public or private IaaS compute cloud.

CloudStack consists of a minimum of two machine installations, in which the primary POD service is called the management server (which manages the entire appliance). The second installation is a hypervisor instance which manages all the cloud services. The hypervisor instance can be one or more nodes and can be part of multi-zone or multi-region deployment.

It can be deployed in some predefined patterns, as listed below.

1. **Small-scale deployment:** This is suitable for test and trailblazer activities.
2. **Large-scale redundant deployment:** Suitable for production-ready (pre-production) and production setups.
3. **Separate storage network:** Used for autonomous storage service on large-scale deployment to manage these services on separate appliances.
4. **Multi-node management server:** Integrated with hardware load balancers for high-availability of services.
5. **Multi-site deployment:** Suitable for enterprise scale deployment with high-performing workloads that have multi-node replication.

For the best solution design, it is better to set up a small-scale deployment solution first for the test environment

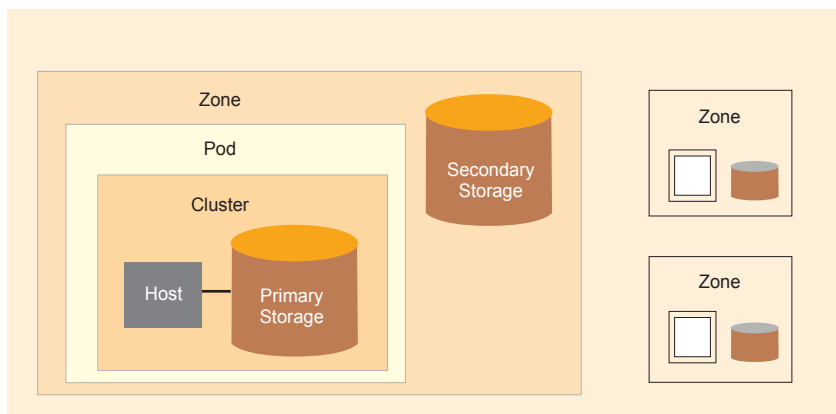


Figure 2: Apache CloudStack architecture